Circumgalactic dust halos update

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October 18, 2023

1 Overview

2 Code flow

I tried to make this a little bit more useable and robust than it was before, interfacing with code via configs, separating runner scripts from class definitions, etc. Also, when creating catalogs, The basic workflow is:

1. Load in the HEALpix masks of foreground and background catalogs.
2. Keep only those objects that lie in the intersection of both HEALPix masks.
3. If needed, join the background redshift catalog with a photometric catalog.
4. If needed, create a foreground random catalog.
5. Save both to file.

2.1 Background catalog creation

2.1.1 Mask handling

2.1.2 Creating background catalogs

The redMaGiC catalog has positions and redshifts, but not magnitudes. To obtain colors, I join them with their counterparts in the Y3 GOLD catalog on the `COADD_OBJECT_ID` parameter. The random galaxy catalog obviously doesn't have a corresponding entry in the GOLD catalog, so instead I match randomly.

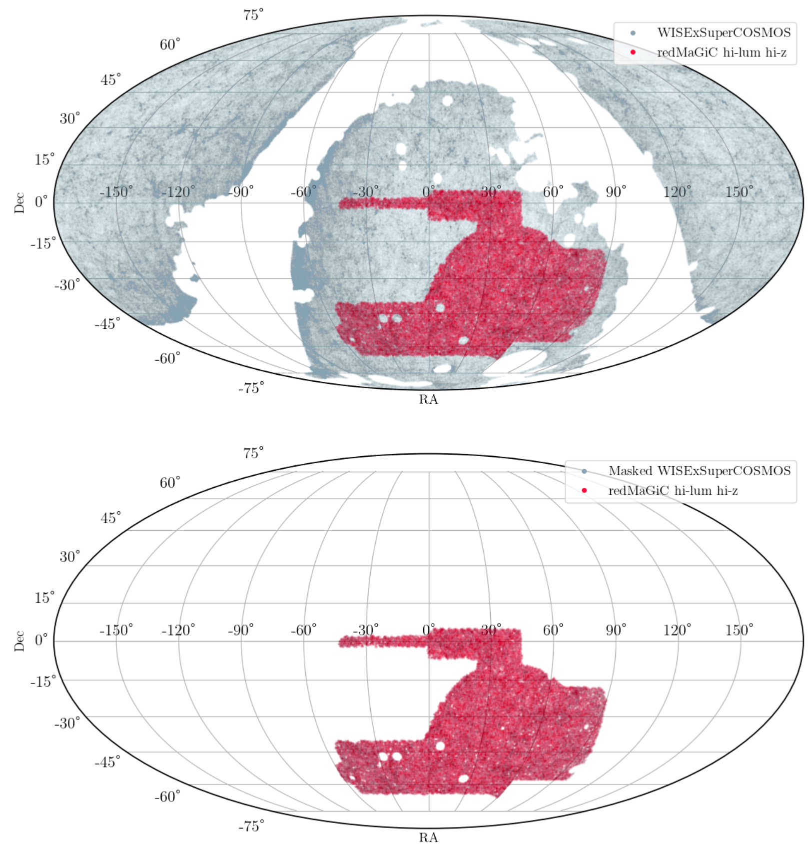

Figure 1: Sky coverage of two catalogs used in this analysis: WISExSuperCOSMOS (foreground, grey points) and the high-luminosity high-redshift redMaGiC galaxy catalog (background, red points). The top row shows the overlap between the full WISExSuperCOSMOS catalog and the redMaGiC catalog; the bottom row shows the overlap of the two catalogs after masking.

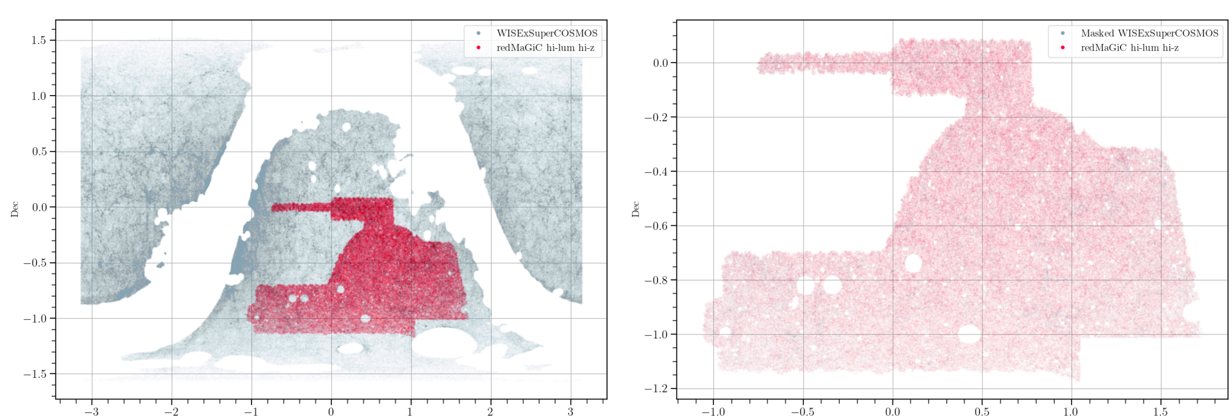

Figure 2: Same as Figure 1, but in rectilinear projection.

Figure 3: Sky coverage of the WISExSuperCOSMOS random galaxy catalog generated following used in this analysis: WISExSuperCOSMOS (foreground, grey points) and the high-luminosity high-redshift redMaGiC galaxy catalog (background, red points). The top row shows the overlap between the full WISExSuperCOSMOS catalog and the redMaGiC catalog with Mollweide (left) and rectilinear (right) projections; the bottom row shows the overlap of the two catalogs after masking.